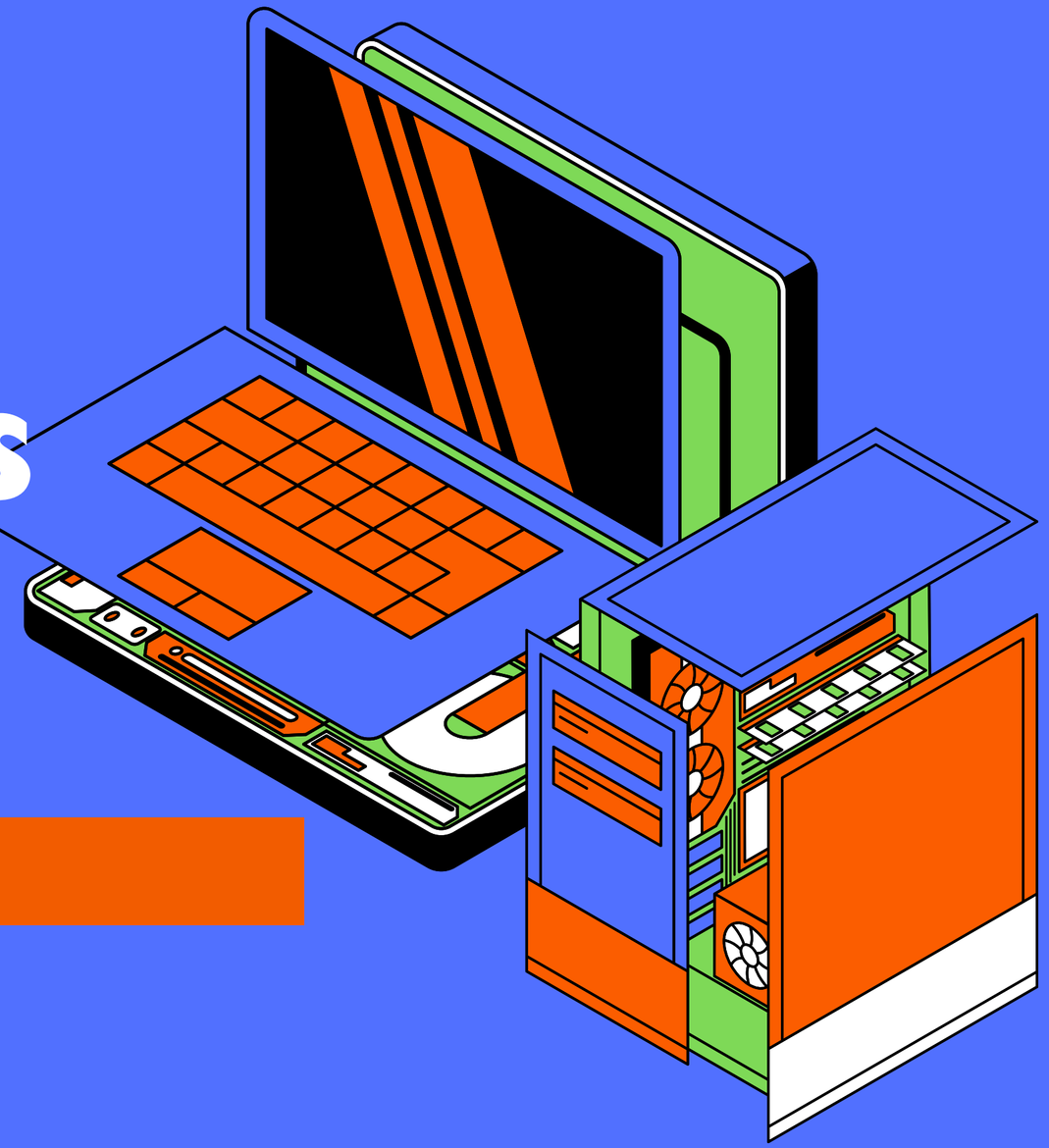
Data Analytics using Python

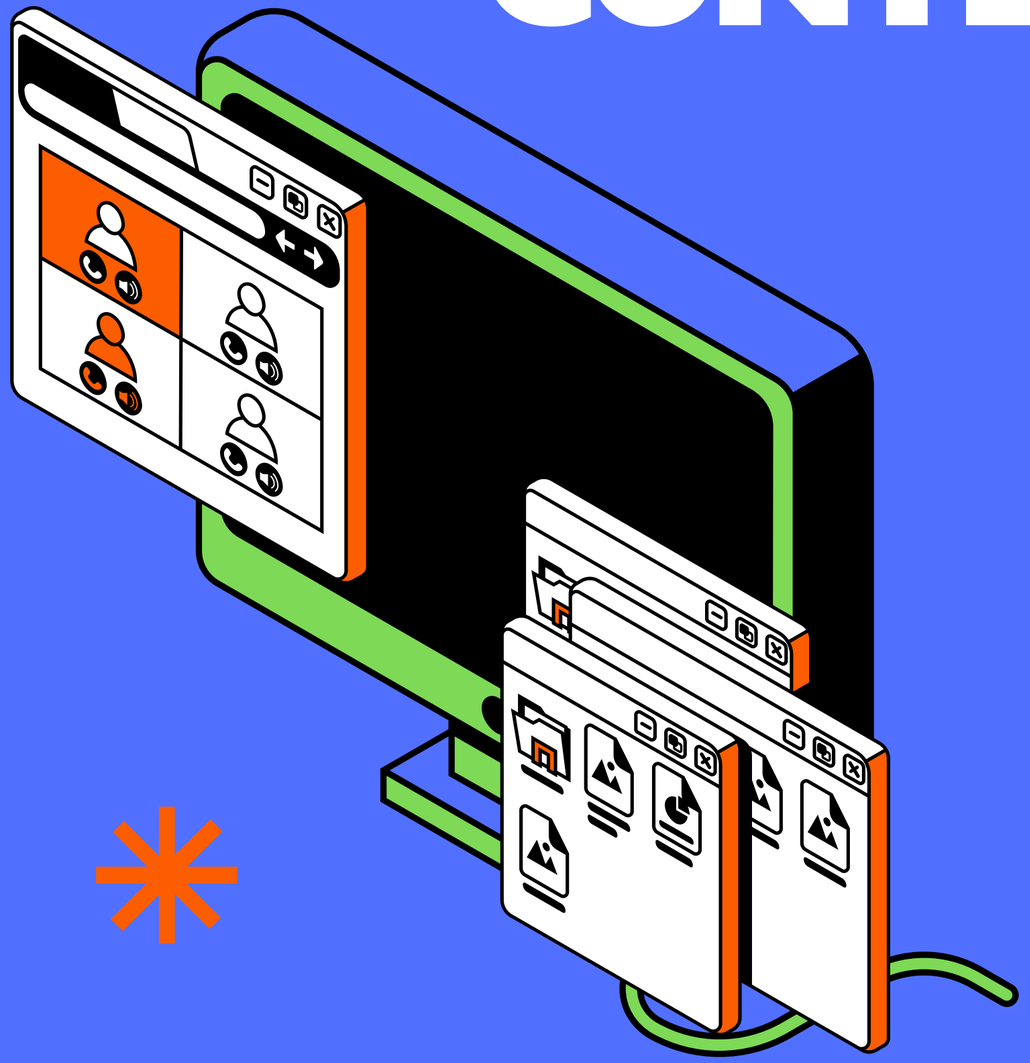
DATA ANALYTICS ON NETFLIX MOVIES & TV SHOWS USING PYTHON



L E A R N I N G T E A M - 2

25PGDM138- Govind Kumar
25PGDM144- Kajol Kumari
25PGDM153- Prajjwal Singh
25PGDM162- Shreya Rai
25PGDM167- Uday Kapoor
25PGDM243- Nikhil Kumar

CONTEXT & PURPOSE



- OTT platforms are highly content-driven
- Netflix operates in a competitive global streaming market

Wrong content decisions lead to:

- High costs
- Low viewer engagement

Purpose of Study:

- To analyze Netflix's content strategy using Python
- To support data-driven managerial decisions

DATASET & TOOLS USED

DATASET:

- NETFLIX MOVIES AND TV SHOWS DATASET
- SOURCE: KAGGLE
- RECORDS: 8,800+ TITLES
- 12 VARIABLES

TOOLS USED:

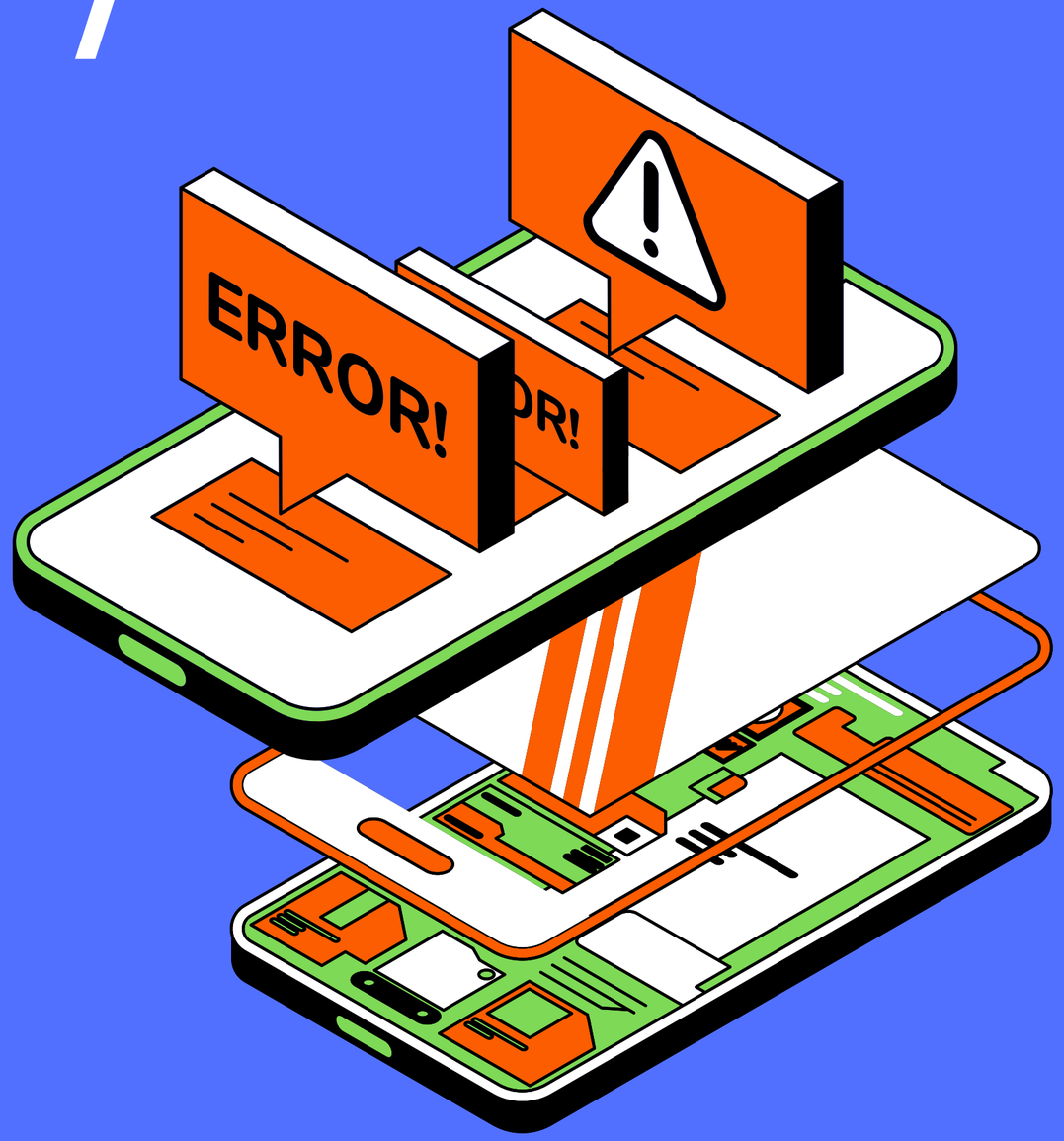
- PYTHON
- PANDAS
- MATPLOTLIB
- SEABORN
- SCIKIT-LEARN



METHODOLOGY / HOW

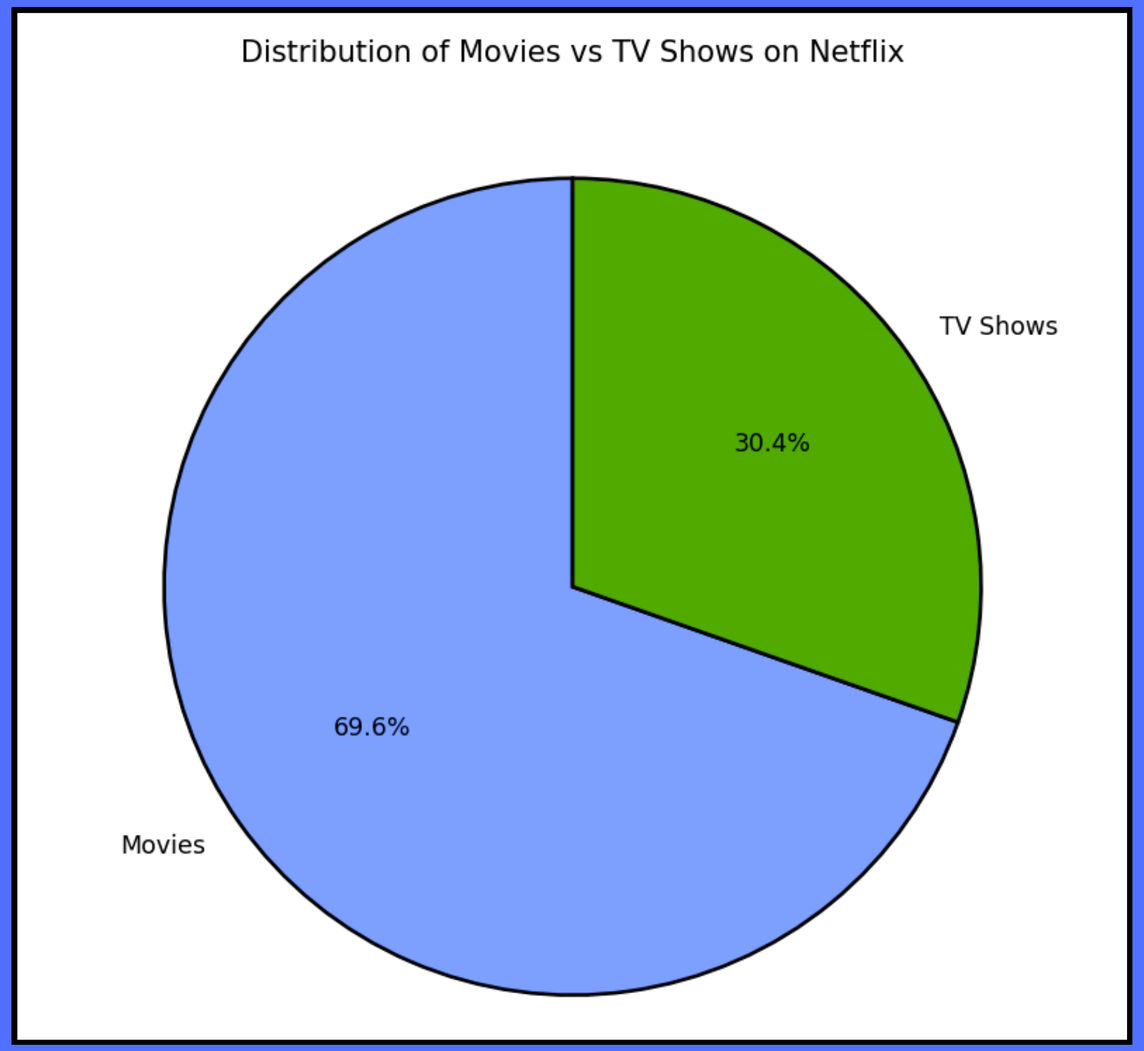
STEPS FOLLOWED:

- LOADED DATASET IN PYTHON
- CLEANED DATASET
- SEGREGATED MOVIES AND TV SHOWS
- PERFORMED 4W + 1H ANALYSIS
- CREATED VISUALIZATIONS
- PERFORMED DESCRIPTIVE STATISTICS
- APPLIED CORRELATION & REGRESSION
- DERIVED BUSINESS INSIGHTS

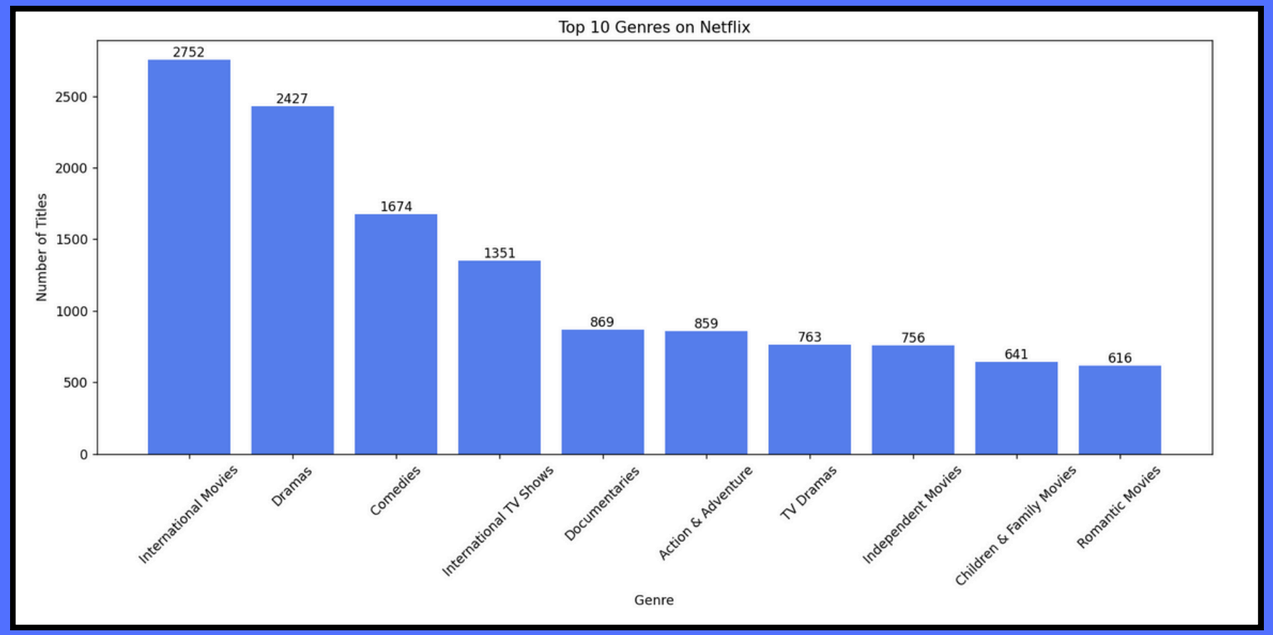


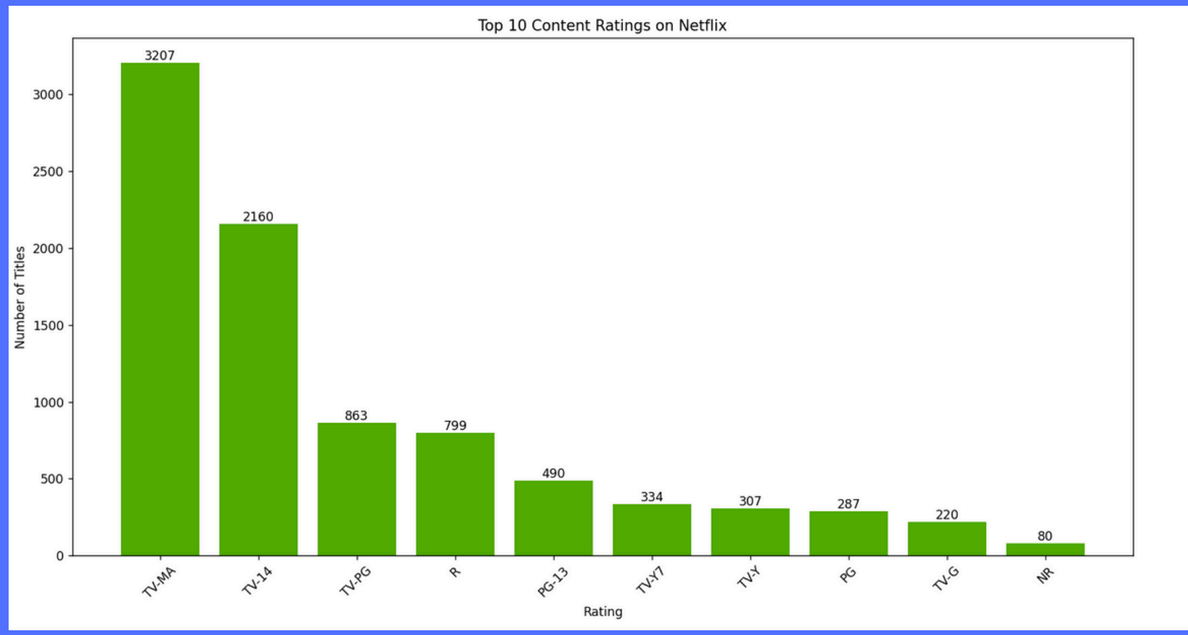
WHAT ANALYSIS (CONTENT MIX)

- Netflix has more Movies than TV Shows
- Indicates preference for short-form content
- Raises question on long-term engagement through series



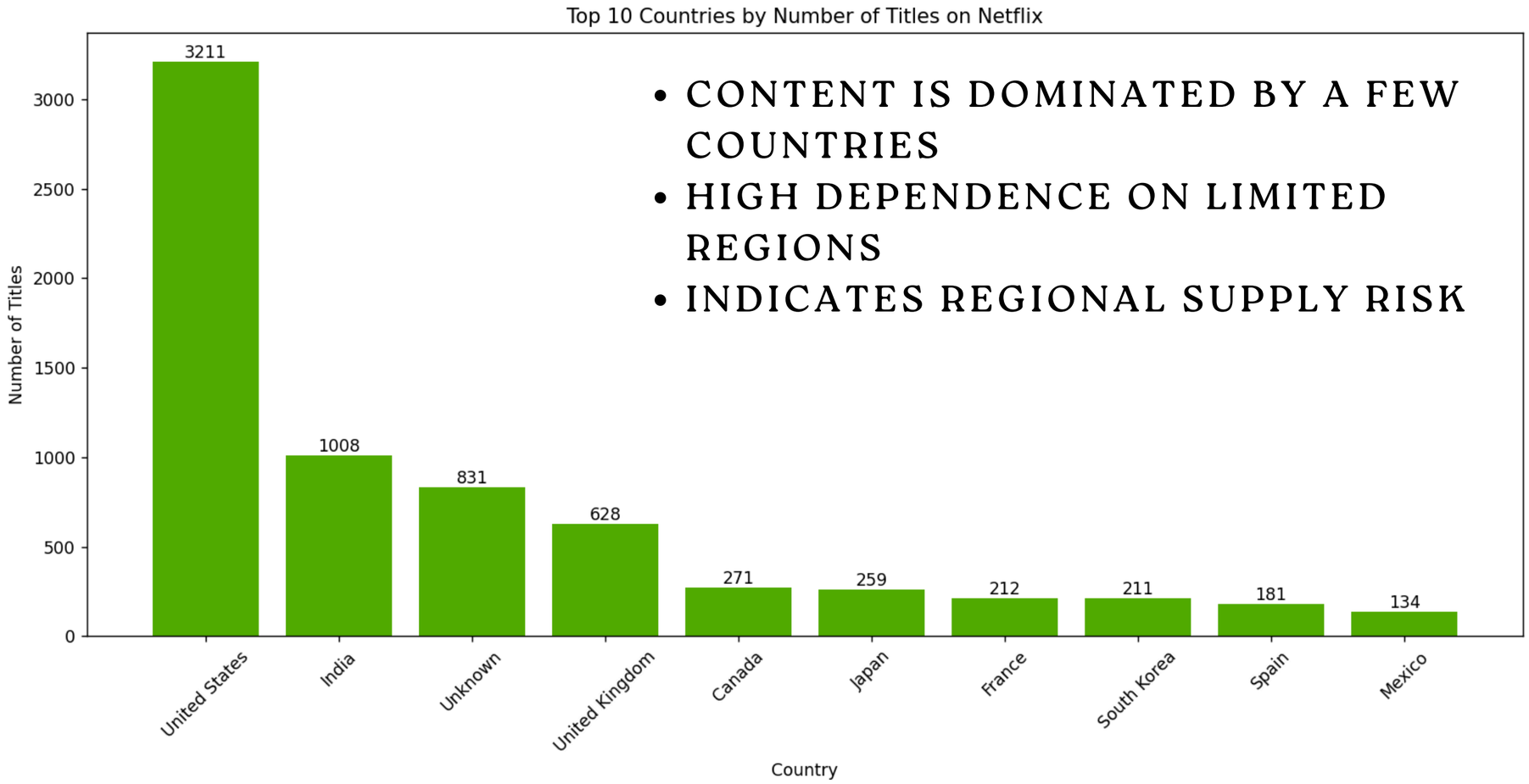
WHAT ANALYSIS (GENRES & RATINGS)





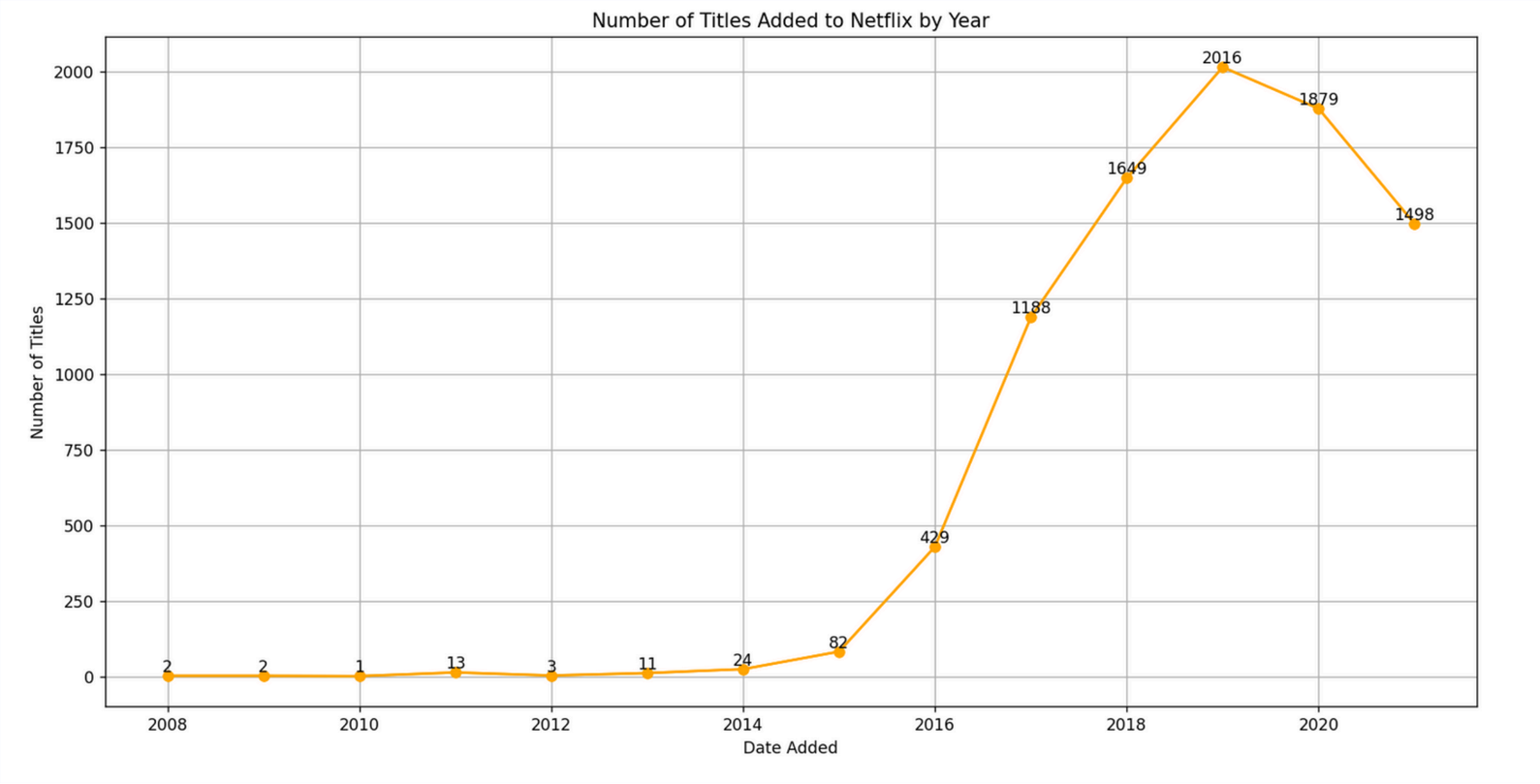
- ONLY A FEW GENRES DOMINATE (DRAMA, COMEDY, ETC.)
- ADULT RATINGS DOMINATE THE PLATFORM
- SHOWS CONTENT CONCENTRATION RISK

WHERE ANALYSIS (GEOGRAPHICAL DISTRIBUTION)



- CONTENT IS DOMINATED BY A FEW COUNTRIES
- HIGH DEPENDENCE ON LIMITED REGIONS
- INDICATES REGIONAL SUPPLY RISK

WHEN ANALYSIS (TIME TREND)



- SHARP RISE IN TITLES AFTER 2015
- REFLECTS OTT INDUSTRY BOOM
- SHOWS AGGRESSIVE CONTENT EXPANSION STRATEGY

WHY ANALYSIS (STRATEGIC REASON)& DESCRIPTIVE STATISTICS

```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```

- MOVIES DOMINATE NETFLIX'S CATALOG

EXPLAINS:

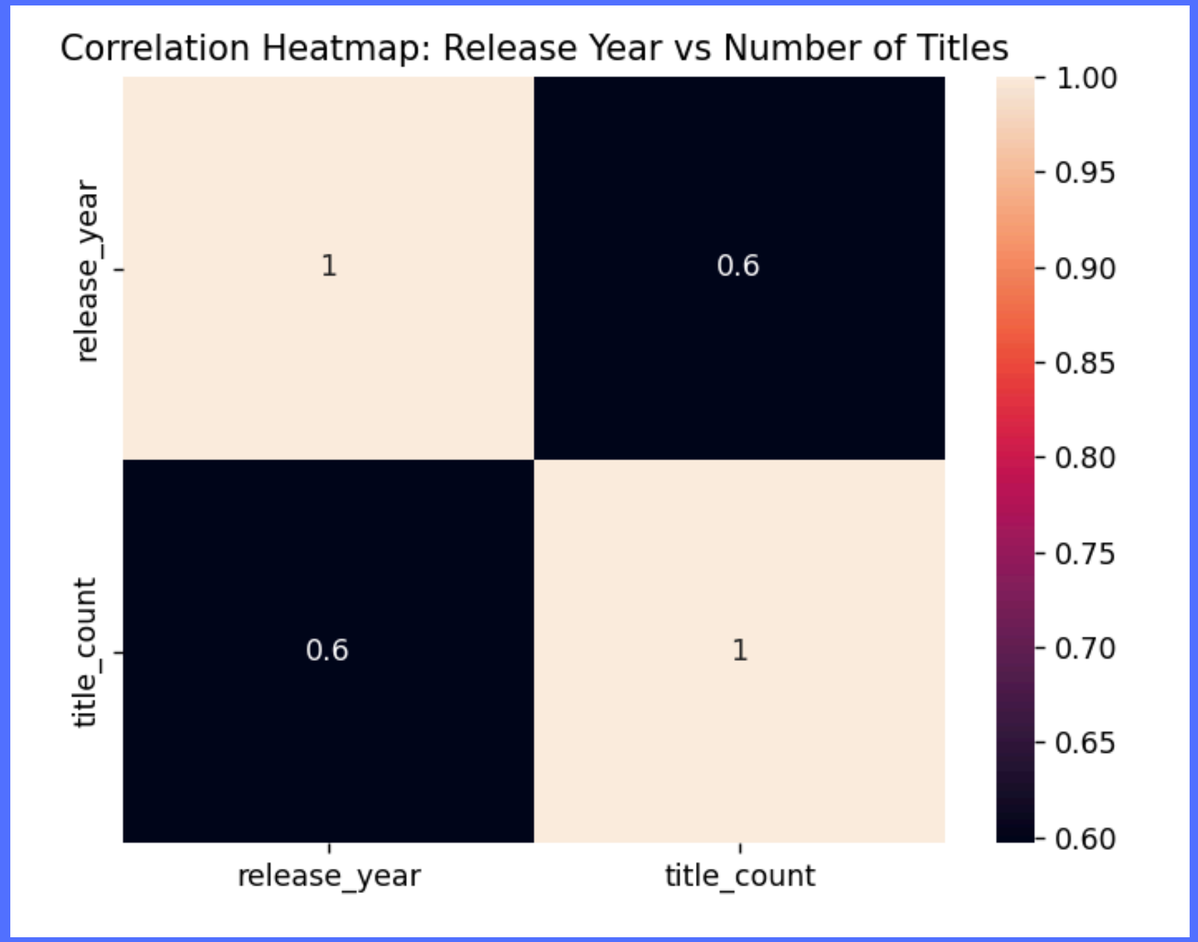
- WHY SHORT-FORM CONTENT IS PREFERRED
- WHY BINGE-BASED STRATEGY IS LIMITED TO CERTAIN CATEGORIES
- SHOWS CONTENT IMBALANCE

```
================== DESCRIPTIVE STATISTICS: Movie Duration =
Most Common Type: Movie
Most Common Rating: TV-MA
Most Common Country: United States
```

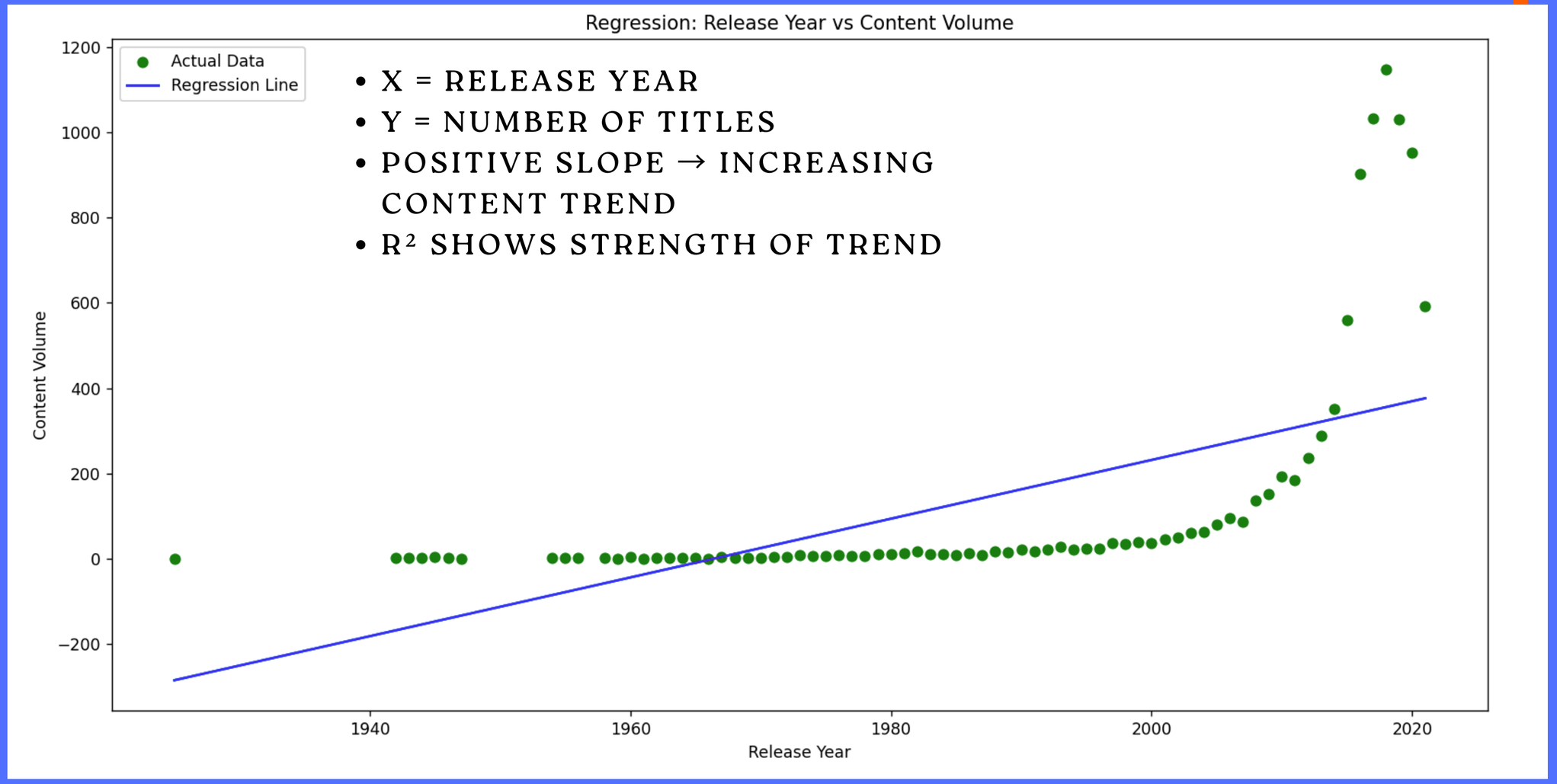
- MEAN & MEDIAN OF RELEASE YEAR SHOW RECENCY OF CONTENT
- MODE SHOWS MOST FREQUENT PRODUCTION YEAR
- HELPS UNDERSTAND CONTENT FRESHNESS

CORRELATION ANALYSIS

- CORRELATION BETWEEN RELEASE YEAR & NUMBER OF TITLES
- POSITIVE RELATIONSHIP OBSERVED
- JUSTIFIES USE OF REGRESSION



REGRESSION ANALYSIS



- X = RELEASE YEAR
- Y = NUMBER OF TITLES
- POSITIVE SLOPE → INCREASING CONTENT TREND
- R^2 SHOWS STRENGTH OF TREND

KEY FINDINGS & MANAGERIAL INSIGHTS

- Netflix is movie-dominated
- Content is genre-concentrated
- Production is geographically concentrated
- Titles increased sharply after 2015
- Strong positive time–content relationship

- Increase TV Show investments for long-term engagement
- Diversify into underrepresented genres
- Expand content production in emerging countries
- Use time trends for budget forecasting

- Python helped convert raw Netflix data into insights
- Content growth is strong but concentrated
- Data analytics supports smarter content strategy

ASSUMPTIONS

ASSUMPTIONS

- Dataset represents Netflix reliably
- First listed country = primary origin
- date_added accurately reflects addition year
- Linear relationship assumed for regression
- Missing fields fixed with “Unknown” do not distort results

In today’s OTT era, content is king — but data is the real crown.

THANK YOU*

LEARNING TEAM - 2

